FML2ILV – Federated Machine Learning

Session 6: Privacy in FL

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Glossary **Book** **GitHub**

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A Simple FL Task

Goal: compute the **average temperature** across n distributed nodes $i = 1, \ldots, n$ on 1-March-2025 in Finland.

- n nodes, each measures local $y^{(i)}$ = temperature
- want the global average $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y^{(i)}$
- each node keeps $y^{(i)}$ private

How to compute $\bar{y}$ collaboratively? $\Rightarrow$ formulate as GTVMin.

Step 1: Formulate as GTVMin

Each node i has scalar model parameter $w^{(i)} \in \mathbb{R}$ and squared-error loss:

$$L_i\left(w^{(i)}\right) = \left(y^{(i)} - w^{(i)}\right)^2$$

$$\boxed{\min_{w^{(1)},\ldots,w^{(n)}} \underbrace{\sum_{i=1}^{n}\left(y^{(i)} - w^{(i)}\right)^2}_{\text{local fit}} + \alpha \underbrace{\sum_{\{i,i'\}\in\mathcal{E}} \left(w^{(i)} - w^{(i')}\right)^2}_{\text{GTV: force agreement}}}$$

Large α: Forced Consensus

As $\alpha \to \infty$, the GTV penalty forces $w^{(1)} = w^{(2)} = \cdots = w^{(n)} := w$:

$$\min_{w \in \mathbb{R}} \sum_{i=1}^{n} \left(y^{(i)} - w\right)^2$$

Solution:

$$\boxed{\widehat{w} = \frac{1}{n} \sum_{i=1}^{n} y^{(i)} = \bar{y} \qquad \text{(the global average)}}$$

- GTVMin with large α and squared-error loss $=$ averaging
- each node contributes its $y^{(i)}$ to the sum

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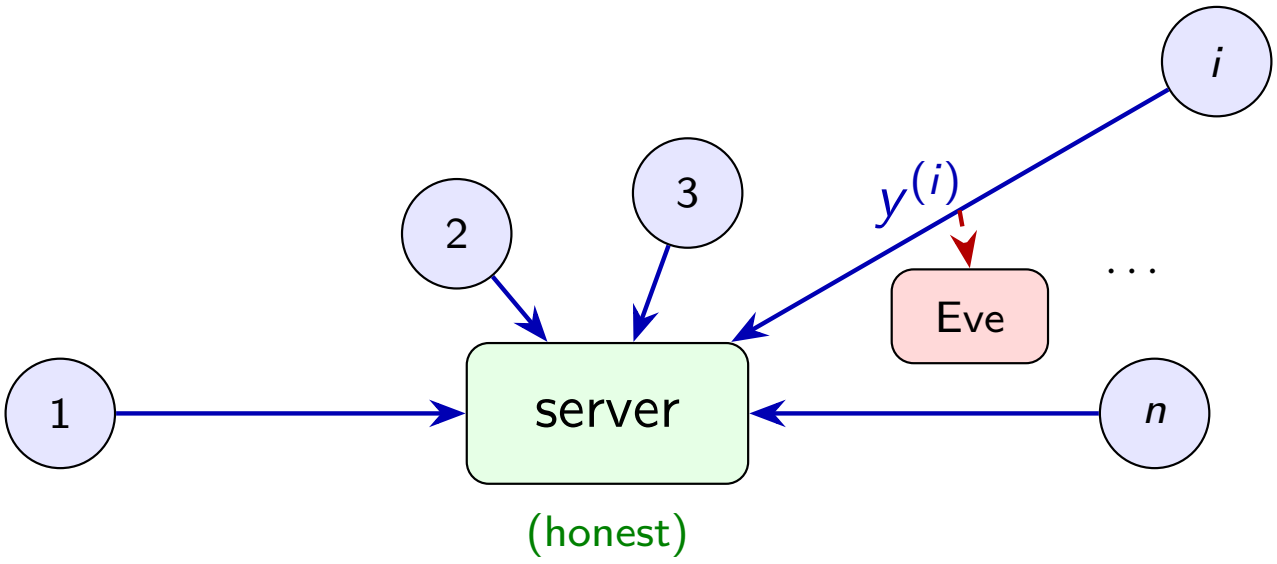
Implementation: Star Network with Eavesdropper

An **honest server** collects measurements to compute $\bar{y}$. An **eavesdropper** (Eve) intercepts node i's communication.



- Eve knows all FMI station temperatures (public data)
- Eve intercepts $y^{(i)}$ from node i
- **can Eve identify which station node i is?**

Temperature Reveals Location

Eve intercepts $y^{(i)}$ and knows all FMI station temperatures.

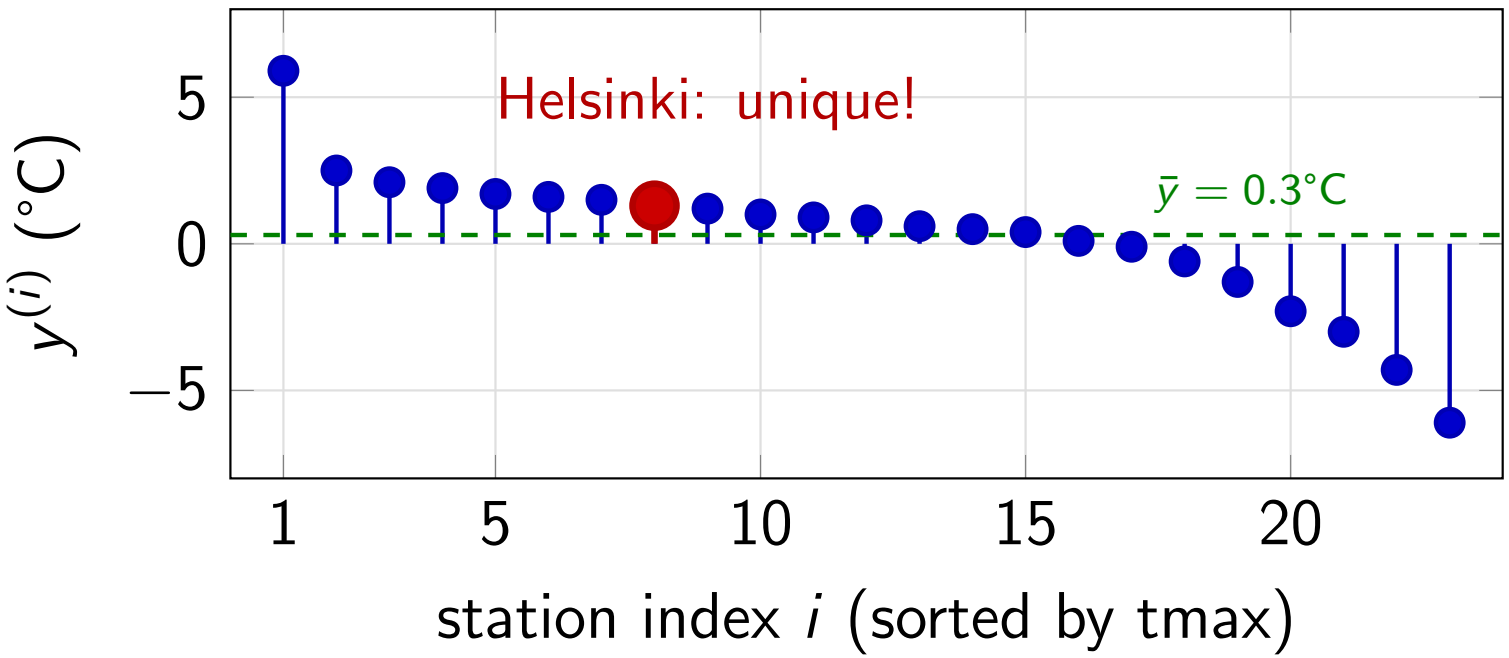
Temperature correlates strongly with latitude in Finland:

- $y^{(i)} \approx 5°\text{C} \;\Rightarrow\;$ Åland islands (Jomala, Lumparland)
- $y^{(i)} \approx 1°\text{C} \;\Rightarrow\;$ southern Finland (Tampere 0.9°C)
- $y^{(i)} \approx -6°\text{C} \;\Rightarrow\;$ far north (Utsjoki)

Eve matches intercepted $y^{(i)}$ against the public FMI temperature list $\Rightarrow$ identifies the exact station and its location.

How Unique is a Single Station?

Eve intercepts $y^{(i)}$ and looks up the public FMI table:



Eve matches intercepted value to the public FMI table $\Rightarrow$ identifies node i's location. Data: `DiffPriv_ToyExample.py`.

Even Gradients Leak Location

Instead of raw $y^{(i)}$, node i sends a gradient to the server:

$$g^{(i)} = -2\left(y^{(i)} - w^{(t)}\right)$$

Eve knows $w^{(t)}$ (it was broadcast by the server), so:

$$y^{(i)} = w^{(t)} - \frac{g^{(i)}}{2}$$

The gradient reveals the local temperature exactly — Eve recovers $y^{(i)}$ and identifies node i's location.

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Idea: Perturb the Gradient

Before sending, node i adds random noise (so Eve sees only a noisy version):

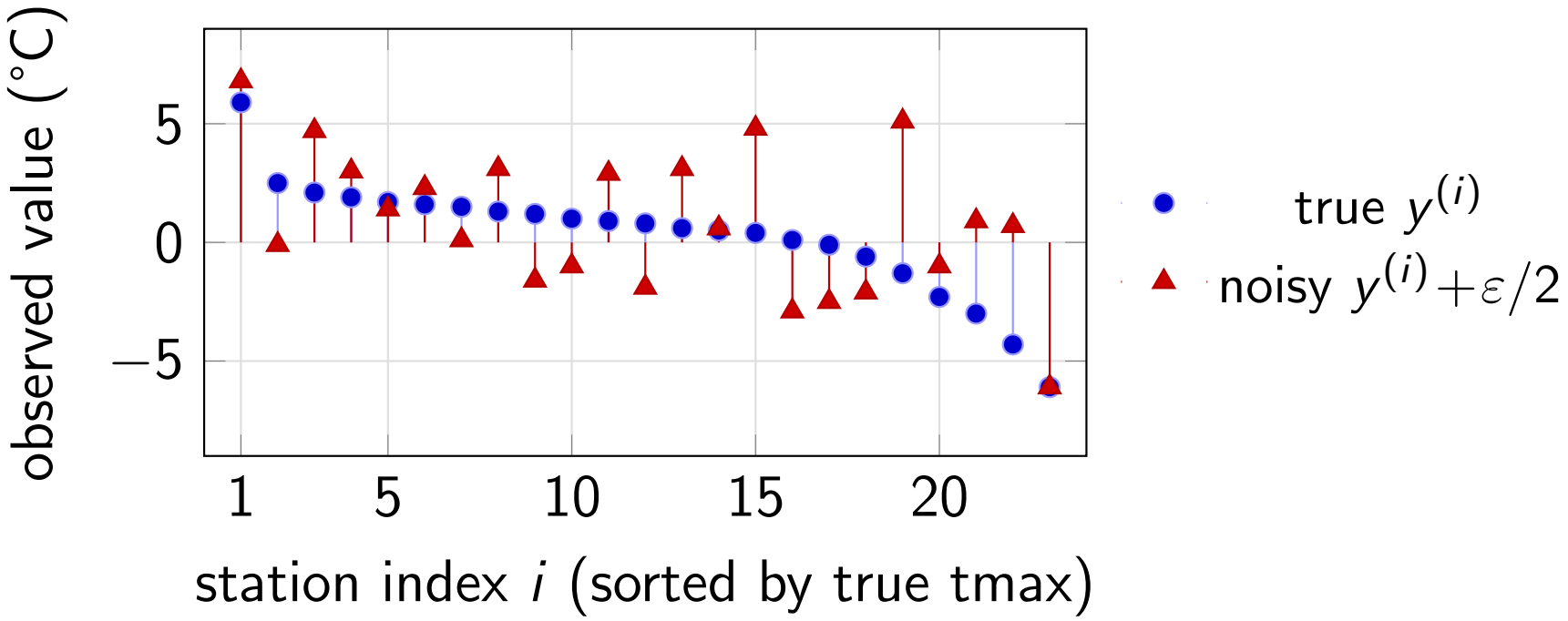
$$\widetilde{g}^{(i)} = g^{(i)} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

Eve now intercepts $\widetilde{g}^{(i)}$ instead of $g^{(i)}$:

$$\widetilde{y}^{(i)} = w^{(t)} - \frac{\widetilde{g}^{(i)}}{2} = y^{(i)} + \frac{\varepsilon}{2}$$

- $\sigma = 0$: Eve recovers exact location (no privacy)
- $\sigma \gg 0$: Eve learns nothing (full privacy, but useless gradient for server)
- moderate σ: **privacy–utility trade-off**

Effect of Noise



With noise ($\sigma = 6$°C), station ordering is scrambled — Eve cannot infer station locations from intercepted temperatures. Data: `DiffPriv_ToyExample.py`.

Eve's Test

Eve knows all FMI station locations and their temperatures (public data).

After intercepting $y^{(i)}$, Eve runs a **hypothesis test**:

H_0**:** node i = Helsinki-Vantaa (tmax $\approx 1.3°$C)
H_1**:** node i = some other station

Without noise: Eve intercepts exact $y^{(i)} = 1.3°\text{C} \Rightarrow$ test is **conclusive** (unique match in the public FMI table)

With noise: Eve intercepts $\widetilde{y}^{(i)} = 1.3 + \varepsilon/2 \Rightarrow$ many stations could have produced this value $\Rightarrow$ test becomes **inconclusive**

Differential Privacy: The Intuition

Goal: make Eve's hypothesis test unreliable — the intercepted value should look *almost the same* regardless of which FMI station is nearest to i.

- H_0**:** node i is near Helsinki (tmax $= 1.3°$C)
- H_1**:** node i is near Tampere (tmax $= 0.9°$C)

If the noisy value looks **nearly identical** in both hypoheses, then Eve cannot tell which station node i is.

Parameter ε controls "how identical": smaller $\varepsilon \Rightarrow$ harder to distinguish $\Rightarrow$ more private.

Gaussian Mechanism

Recipe: add Gaussian noise calibrated to the **sensitivity** Δ:

$$\widetilde{g} = g + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2)$$

- Δ = max change in output when one station's input changes
- noise: $\sigma \ \propto \ \Delta/\varepsilon$ (more sensitive $\Rightarrow$ more noise needed)
- ε: privacy budget — smaller = more private, more noise

Sensitivity of the Individual Transmission

Eve intercepts the **single value** $\widetilde{y}^{(i)} = y^{(i)} + \varepsilon_i$ from node i.

Sensitivity: how much can $\widetilde{y}^{(i)}$ change if $y^{(i)}$ changes?

$$\Delta = B \qquad \text{(full temperature range, e.g. } B = 12^\circ\text{C)}$$

Note: $\Delta = B$, **not** B/n. We protect the *individual transmission* that Eve intercepts, not the average that the server computes.

Gaussian mechanism requires each node to add noise:

$$\sigma = \frac{B\sqrt{2\ln(1.25/\delta)}}{\varepsilon}$$

Utility of the Noisy Average

The honest server averages n independently noisy values:

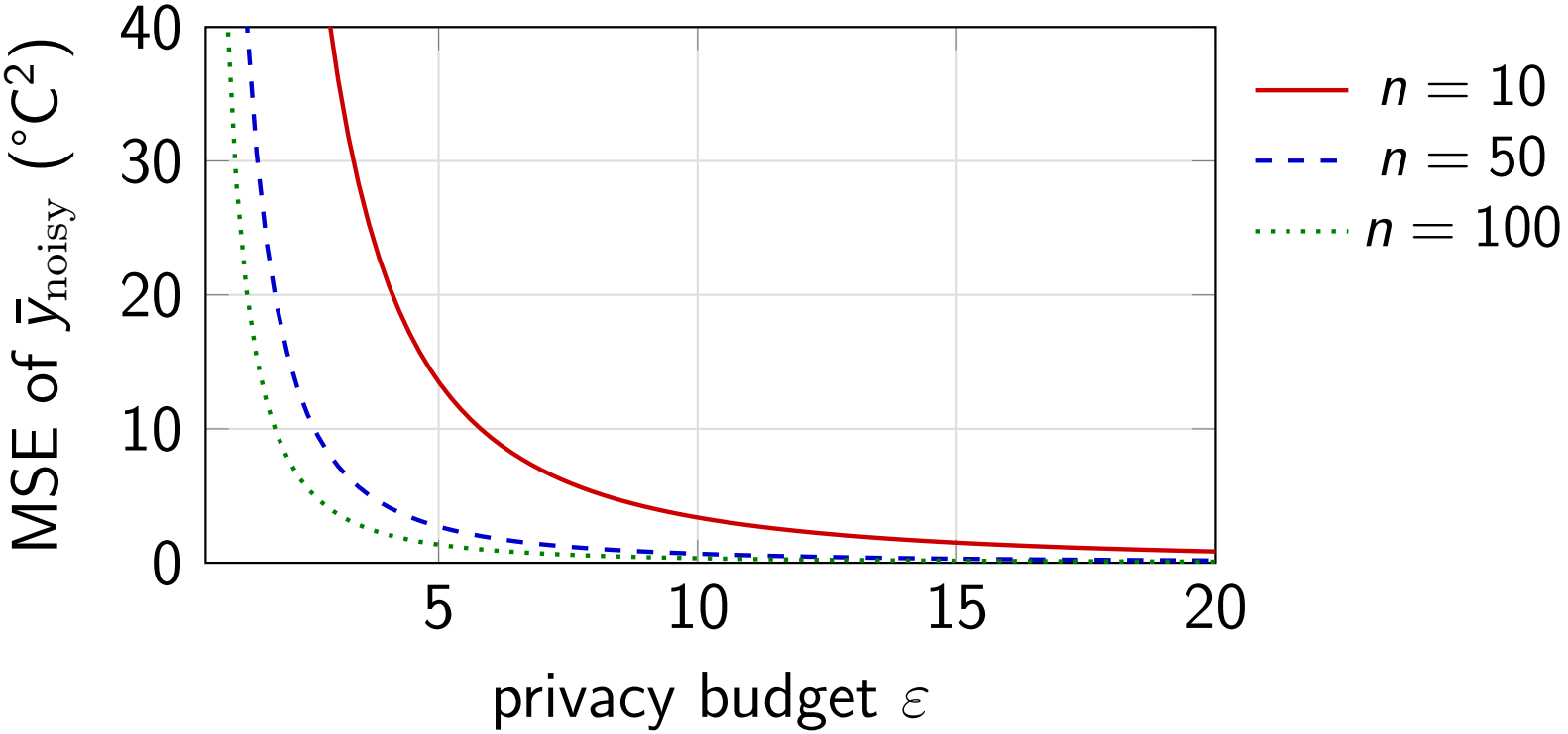
$$\bar{y}_{\text{noisy}} = \frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} + \varepsilon_i\right) = \bar{y} + \frac{1}{n}\sum_{i}\varepsilon_i$$

Mean squared error of the noisy average:

$$\boxed{\mathrm{MSE} = \frac{\sigma^2}{n} = \frac{B^2\, 2\ln(1.25/\delta)}{n\,\varepsilon^2}}$$

- more nodes $n \uparrow \Rightarrow$ MSE $\downarrow$ (**privacy amplification**)
- stronger privacy $\varepsilon \downarrow \Rightarrow$ MSE $\uparrow$
- the noise "averages out" across nodes — but each node must still add σ

Privacy–Utility Trade-off



$B = 12$°C, $\delta = 10^{-5}$. More nodes $\Rightarrow$ curve shifts down (same privacy, better accuracy). At $n = 100$, $\varepsilon = 5$: MSE ≈ 1.4°C^2.

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- **Averaging = GTVMin** with squared-error loss and $\alpha \to \infty$
- **Gradient leakage:** a single gradient reveals local measurement exactly
- **Defence:** add calibrated noise (DP)
- **Trade-off:** ε controls privacy vs. utility; more devices $\Rightarrow$ less noise needed